

Writing and Solving Equations in Real-World Contexts

Additional Practice 7A.01.07 **ANSWER KEY**

For questions 1 through 6, define the variable, write an equation, solve it, and say what the solution means.

1. Marcus buys five identical gift boxes online and pays a \$6.95 shipping charge on the order. The total came to \$91.70. What does one gift box cost?

Variable. $b = \text{the cost of one gift box.}$

Equation. $5b + 6.95 = 91.70$

$$5b = 84.75$$

$$b = 16.95$$

Check: $5(16.95) + 6.95 = 84.75 + 6.95 = 91.70 \checkmark$

Meaning. $\text{One gift box costs \$16.95.}$

Shipping is charged once on the whole order, so it is outside the group.

2. A bakery sold 12 fewer loaves today than twice the number it sold yesterday. Today it sold 5 times as many loaves as the café next door, which sold 8. How many loaves did the bakery sell yesterday?

Variable. $y = \text{the number of loaves sold yesterday.}$

Equation. $2y - 12 = 40$

The café sold 8, so the bakery sold $5(8) = 40$ loaves today.

$$2y = 52$$

$$y = 26$$

Check: $2(26) - 12 = 52 - 12 = 40 \checkmark$

Meaning. $\text{The bakery sold 26 loaves yesterday.}$

Find the 40 before writing the equation.

3. Jamal has a \$75 gift card. He buys six identical items and has \$13.50 left on the card. What did each item cost?

Variable. $x = \text{the cost of one item.}$

Equation. $75 - 6x = 13.50$

$$-6x = -61.50$$

$$x = 10.25$$

Check: $75 - 6(10.25) = 75 - 61.50 = 13.50 \checkmark$

Meaning. $\text{Each item cost \$10.25.}$

4. A summer camp charges each of seven children a \$22 registration fee plus the same activity fee. The seven families paid \$364 in all. What is the activity fee?

Variable. $a = \text{the activity fee for one child.}$

Equation. $7(22 + a) = 364$

$$22 + a = 52$$

$$a = 30$$

Check: $7(22 + 30) = 7(52) = 364 \checkmark$

Meaning. $\text{The activity fee is \$30 per child.}$

Each child gets the same pair of charges, so the group repeats and the parentheses belong there.

5. A granola recipe uses $\frac{2}{5}$ cup of oats for each serving, plus 1.4 cups of oats for the topping. The whole batch uses 7 cups of oats. How many servings does the recipe make?

Variable. $s = \text{the number of servings the recipe makes.}$

Equation. $\frac{2}{5}s + 1.4 = 7$

$$\frac{2}{5}s = 5.6$$

Multiply by the reciprocal: $s = 5.6 \times (5/2) = 14$

Check: $0.4(14) + 1.4 = 5.6 + 1.4 = 7 \checkmark$

Meaning. $\text{The recipe makes 14 servings.}$

$2/5 = 0.4$, so dividing by 0.4 works too and gives the same 14. The reciprocal route is faster on paper.

6. A gas bill charges \$0.98 for each therm used plus a \$16.50 service charge. Last month's bill was \$114.50. How many therms were used?

Variable. $n = \text{the number of therms used last month.}$

Equation. $0.98n + 16.50 = 114.50$

$$0.98n = 98$$

$$n = 100$$

Check: $0.98(100) + 16.50 = 98 + 16.50 = 114.50 \checkmark$

Meaning. $100 \text{ therms of gas were used last month.}$

Question 7 asks whether the answer makes sense. Question 8 is error analysis.

7. A field trip costs \$260 plus \$17 for each student. The budget is \$700. How many students can go?

Variable. $s = \text{the number of students who can go.}$

Equation. $260 + 17s = 700$

$$17s = 440$$

$$s = 25.88\dots$$

25 students can go.

Check: $260 + 17(25) = 260 + 425 = \685 , and 26 students would cost \$702.

Meaning. *The trip can take 25 students and stay inside the \$700 budget.*

Round down. The equation allows 25.88 students; the \$700 budget does not — 25 students cost \$685, and 26 would cost \$702.

8. Rosa was solving $\frac{1}{8}s + 0.75 = 2$. She wrote $\frac{1}{8}s = 1.25$ and then $s = 1.25 \div 8 = 0.15625$. Explain her error and give the correct solution.

Her first line is correct. The error is in the second.

Multiplying by $\frac{1}{8}$ is the same as dividing by 8. The inverse of dividing is multiplying, so she should have multiplied by 8, not divided again.

$$s = 1.25 \times 8 = 10$$

Check: $\frac{1}{8}(10) + 0.75 = 1.25 + 0.75 = 2 \checkmark$

Math Nation names this exact error in its own misconceptions list for this lesson: students try to divide by 8 to undo multiplying by $\frac{1}{8}$. The reasonableness check is against her own correct second line: $(\frac{1}{8})s = 1.25$ says one EIGHTH of s is 1.25, so s has to be EIGHT TIMES 1.25. Her 0.15625 is 1.25 divided by 8, which is 64 times too small.

Questions 9 and 10 look ahead to multi-step equations, which arrive later this year. The unknown appears on both sides of the equal sign.

9. Solve $4(n + 5) = 6n - 2$. Check your solution by substitution.

Equation. $4(n + 5) = 6n - 2$

$$4n + 20 = 6n - 2$$

$$20 = 2n - 2$$

$$22 = 2n$$

$$n = 11$$

Check: $4(11 + 5) = 4(16) = 64$, and $6(11) - 2 = 66 - 2 = 64 \checkmark$

Meaning. $n = 11$ makes both sides equal 64.

Distribute first. Moving the smaller n term keeps the coefficient positive.

10. One tank holds 240 gallons and is draining at 8 gallons a minute. A second tank holds 60 gallons and is filling at 12 gallons a minute. After how many minutes do the two tanks hold the same amount?

Variable. $t = \text{the number of minutes since the tanks started.}$

Equation. $240 - 8t = 60 + 12t$

$$240 = 60 + 20t$$

$$180 = 20t$$

$$t = 9$$

Check: $240 - 8(9) = 168$, and $60 + 12(9) = 168 \checkmark$

Meaning. *After 9 minutes both tanks hold 168 gallons.*

Adding $8t$ to both sides turns the negative rate into a positive term. One tank is going down and the other up, so they must meet.